

Technology Stack

Date	27 June 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	2 Marks

Define Your Technology Stack:

The project uses a standard model-view-controller (MVC) inspired layout built on Flask, bridging Python-based machine learning models with a responsive web browser UI. architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3	Provides a responsive, userfriendly interface for applicants and bank officials to enter loan details and view prediction results

2	Backend / Server-Side	Python, Flask	Flask handles HTTP requests, integrates the trained machine learning model, processes applicant data, and returns loan approval predictions efficiently.
3	Database / Data Storage	SQLite(If it requires)	Stores applicant details, prediction results, and loan history securely while supporting efficient data retrieval.
4	Cloud / Hosting / Deployment	IBM Cloud	Hosts the Smart Lender application, enabling scalable, reliable, and cloud-based deployment for real-time loan prediction.

5	Version Control & CI/CD	Git, GitHub	Supports source code management, team collaboration, version tracking, and simplifies project maintenance
6	Third-Party APIs / Other Tools	Scikit-learn, XGBoost, Pandas, NumPy, Joblib	Scikit-learn and XGBoost are used to build and evaluate ML models. Pandas and NumPy handle data preprocessing and analysis. Joblib is used to save and load the trained XGBoost model for deployment.